Voiland Food Pantry

Final Report Draft 2

Mentor: Maynard Poleng Siev, WSU

Team Name: 11-FA25-SP26-F-MARKETING

Members: Jodie Butterworth

CptS 423 Software Design Project II

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Link to Video for Sprint5: https://youtu.be/85BKI1TZN80

Link to GitHub: https://github.com/WSU-JB5757/WSU_Marketing_423

I. Introduction

As stated in the promotional material*, Washington State University, in an effort to reduce food insecurity, has opened food pantries on campus to provide free food. The development team has created a program to read card data for volunteers and beneficiaries, and provide a database to track the pantry inventory. In addition to the Voiland Food Pantry, the program has potential uses at other campus facilities. The sponsor has requested a rollout plan including all necessary technical documents, testing and specifications of the current program to prepare it for scaling, maintenance, distribution and future revisions by other facilities and development teams. The software program will be tested, documented and prepared for deployment with documentation and revisions from the product development team. The marketing team does not provide code or product development for future program revisions. Instead, it organizes information and plans goals for product rollout and future development.

II. Team Members & Bios

II.1. Jodie Butterworth

Jodie is a senior Computer Science major at Washington State University. Her skills include UI/UX mobile and web development, graphic design, and technical writing. She has prior experience with web development and has worked with WordPress, as well as programming languages such as Python, C/C++/C#, HTML, and CSS. For this project, her responsibilities include front-end development and UI design.

As described in earlier Project Drafts**.

* Introduction from the promotional material abstract for the poster presentation.

** Bio from the team bio section of the development team project report dated 12-7-25.

III.Project Requirements Specification

As agreed in the Client meetings, the scope of the rollout for the marketing project is:***

- Create timeline for product deployment with multiple locations and stakeholders.
- Write a comprehensive user guide for admins, future clients and future teams that includes troubleshooting common problems.
- Create a plan for continued maintenance including hand-off to future project teams.
- Plan marketing meeting and hands-on demonstration with potential future clients.
- Identify potential costs and fees of operating the system.
- Identify quality and performance issues with testing, implement repairs, and record the process and results.
- Create a user's guide for practical daily operation.
- Perform risk analysis on hardware and data secure and notify development team for adjustments.
- Develop a step-by-step plan to save program, data and databases for re-deployment on different hardware.

III.1. Project Stakeholders

List of Stakeholders:

- Clients for the Voiland Food Pantry who want a rollout plan for the scanning software to manage inventory and card readers to manage client and volunteer data
- Potential clients in other departments who want to improve their inventory, volunteer and employee data management
- Future users, team project developers and administrators who want clear documentation of the program and how to use and troubleshoot it
- Current development team who want to develop the project with minimal issues.
- Future program users, volunteers and clients who want an easy-to-use program with risk analysis, bug testing, and clear user guides and documentation.

*** As described the meeting minutes for the meeting on 3-5-26, and adjusted in the meeting minutes on 3-26-26

III.2. Use Cases

- U1 As a volunteer of the Voiland Food Pantry, I want to read the user manual so that I know how to run the terminal.
- U2 As an admin of the Voiland Food Pantry, I want to read the comprehensive user guide so that I know how to change the program title.
- U3 As a developer on a future improvement to the program, I want to look at the comprehensive user guide so that I know how to migrate the program and data to different hardware.
- U4 As a new client I want to read the comprehensive user guide so that I know how to operate the database.
- U5 As a developer of a future improvement to the program, I want to read the testing documentation so that I understand the program operation and the testing used.

III.3. Functional Requirements

Based on the objectives from client meetings:***

- FR1 Development team to produce user guide for daily operations that will be edited by marketing.
- FR2 Development to provide information to marketing for a comprehensive user guide for future admins and project teams.
- FR2 Development team to write a report on any costs or fees in operating the product and any foreseen costs of fees of future operations.
- FR3 Development team to provide information to marketing to produce technical specifications and a migration plan.
- FR4 Development team to provide information to marketing to create a list for troubleshooting common problems.
- FR5 Development team to record procedures and results of testing processes and the repairs implemented.

III.4. Non-Functional Requirements

Based on the discussion from client meetings:***

- NR1 Improve communications with development team.
- NR2 Teams to be used in timeline planning.

- NR3 Meetings to be held with potential future clients.
- NR4 Client to be CC in all email communications for the project.
- NR5 Documents to be edited and easy to use.
- NR6 Risk analysis performed on terminal hardware and software in situations of low-supervision and customer self-service.
- Potential user and client feedback to be gathered at demonstration meeting

III.5. Standards

This project is related to a software design project, but is itself not software based. Software testing will be done and compared with other testing results. Teams will be used as project management and communication software. Email is the main communication method.

I am planning to implement AI-based project management to create a more complete and interactive rollout planner. It will be timeline-based with email reminders and appropriate forms attached to events. It will be rather basic, and likely python-based with connection to LLM AI through an API key. Teams and email will still be used as communication. GitHub will still be used as storage and to turn things in. This will not be a large program, and it will be based on a YouTube or internet tutorial series.

Aside from that, the rollout documentation mush be edited, complete, easy to use and contain all the required information. Meetings with the client and the instructor may alter the goals as the project moves forward.

IV. Software Design - From Solution Approach

Currently, cloud-based hosting is being analyzed by the development team as a cost-effective deployment alternative for future clients and teams. The Firefox-based UI is facing technical issues, and is being repaired by the development team. As of this writing, these are the only technical issues from testing.

After risk analysis of the operating environment, additional encryption will be added to the card reader data, and the physical hardware will be secured from theft and tampering at the terminal. This is to enable the Voiland Food Pantry to operate safely even when a volunteer is not monitoring the pantry.

For the AI portion, it will be a better timeline organizer with more features for communication and AI assistance incorporated. It will help create emails and forms and email reminders. It will be based on available tutorials, and it will rely on external AI sites.

IV.1. Architecture Design

The real program is made by the development team for this project.

The smaller AI project will be in python, and will connect email and LLM AI to a timeline with data to process into emails and forms. I may add diagrams and descriptions about it, but the real project code for this marketing project is in the related development team project.

IV.2. Data design

The real program is made by the development team for this project.

Since it’s a small program, the instructional data will likely be in the python program. Since it’s connected to AI, it will pull other information from online AI.

IV.3. User Interface Design

The real program is made by the development team for this project.

I will use a python interface for simplicity. I may add a GUI if time allows.

IV.4. Constraints

See the project scope under Project Requirements.

V. Test Case Specifications and Results

V.1. Testing Overview

Testing has been conducted by the development team. The only major errors were in the Firefox UI. User testing continues in the FIZ under the client’s supervision. Initial reports are that it is easy to use as a self-service terminal, and has no other major errors.

From a marketing project perspective, the documentation was only a user guide. It did provided enough features to describe basic program operation as a client, but not enough for admins or future teams. It lacked a good workaround for the Firefox issue, so unfortunately, the documentation is not comprehensive enough for a user self-service document.

V.2. Environment Requirements

The Sloan Building has experienced a string of thefts recently. The terminal should not be left alone and unsecured. Testing of the software continues, and a plan has been developed to physically secure the terminal for future self-service operation.

V.3. Test Results

Testing found errors in the UI, but the rest of the program is fully operational for customers. The Firefox-based UI does not always load correctly. The user guide is not comprehensive enough to include operation with this known issue. Therefore, user case U1 is not currently satisfied.

The comprehensive guide for the admins and future development teams is not yet ready, and testing of the program as admins has not yet been conducted.

VI. Projects and Tools used

I'm using GitHub, Teams, Python and word processors.

VII. Description of Final Prototype

The final project will include a user manual, stakeholder feedback forms, a comprehensive user guide for admins and future teams that includes troubleshooting and editing the program and database, an analysis of operating costs for future operations, testing and testing records, a timeline with tasks for the rollout and an AI planner to automate some of these tasks.

VIII. Social Responsibility and Broader Impacts

The food pantry provides a valuable community service. Improving their software to more efficiently track volunteer hours, customer use, and product inventory will enable them to do their job more efficiently. A rollout plan makes the transition for the next team who takes over with the project improves easier for them. It also connects with potential clients and provides both current and future clients and administrators the records and instructions they need to better utilize the program. It also analyzes the environment for risks and warns the development team of potential issues.

IX. Product Delivery Status

A booklet for the poster presentation was developed, but it has been merged with the development team presentation.

Initial testing is completed and error-correction for the UI by the development team has begun.

The cost-analysis and resulting planning for migration of the database to cloud or other hardware is also being planned by the development team.

The development team sent a simple user guide for testing purposes. The user guide was forwarded to the client for review, and used in testing.

A rollout timeline has also been established on Teams, as per client request.

Environment-based risk analysis done and the development team was notified of potential risks. Plans to further encrypt data and physically secure the terminal have been made.

Plans for potential client meeting with stakeholder feedback and project demonstration set for early April.

X. Conclusions and Future Work

X.1. Limitations and Recommendations

I do not directly contribute code for the development team. They are responsible for giving me information to make the comprehensive user guide and stakeholder feedback form. They are creating a plan to migrate the program and host it on the cloud or other hardware. They are also creating modifications to the UI. As a note, I was involved in creating the prototype and in the project planning, but I no longer contribute code or plans.

X.2. Future Work

I will collect program and testing information and create the comprehensive user guide and the stakeholder feedback form. The future deployment of the program to other departments and locations will be partially dependent on this meeting and demonstration. The rollout plan will adjust based on known future plans.

The AI planner adds code to this side project, but is not the proper Voiland Food Pantry project code. I will continue to use email, Teams and GitHub with the planner as an experimental addition, really.

XI. Acknowledgements

I thank Maynard Poleng Siev for sponsoring the marketing project. I thank the development team and my instructor for reviewing the marketing project and for putting in the extra work to make it viable.